QM1 Statistics: week 4

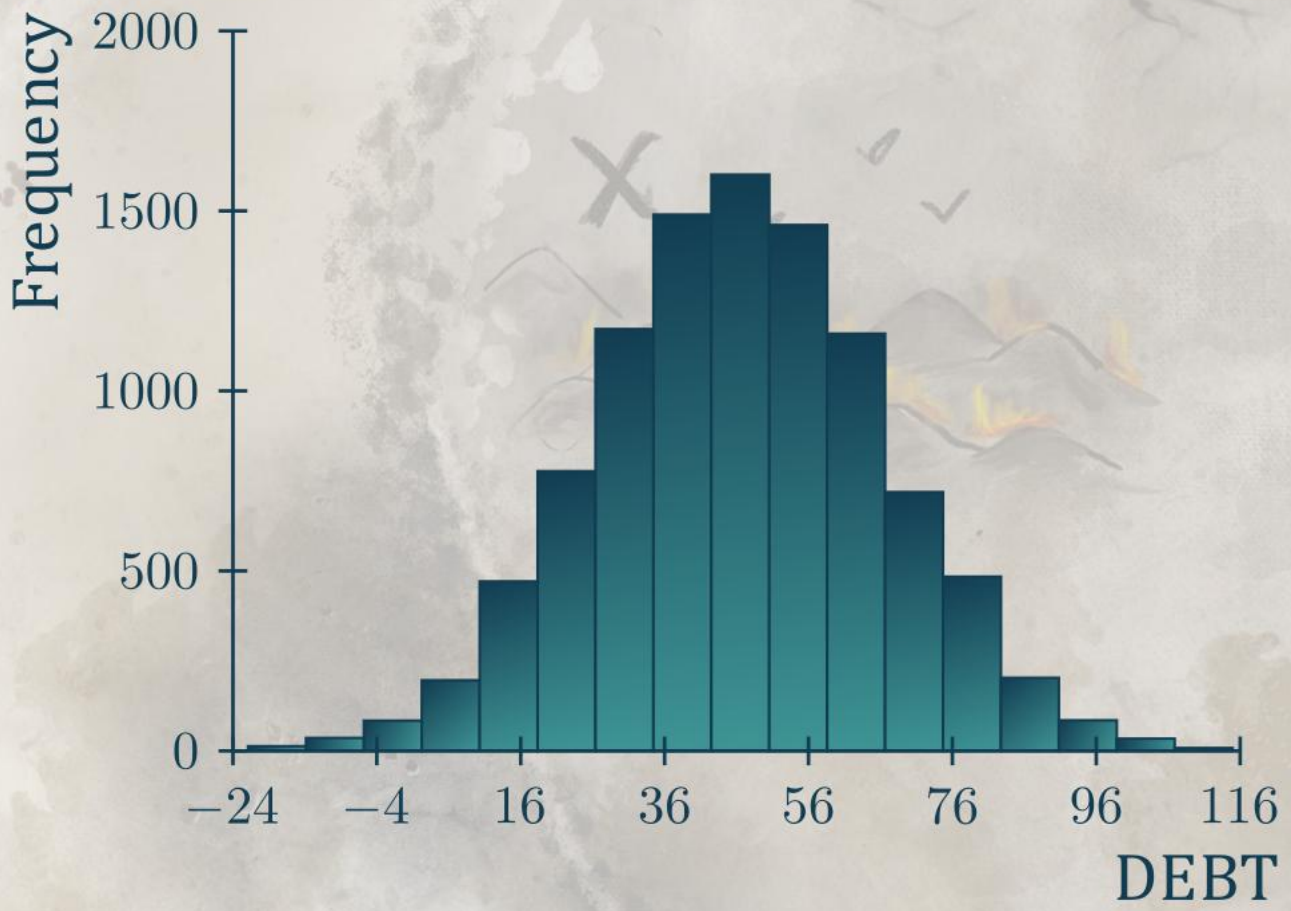
The Normal Distribution and Sampling Designs

Welcome!

Normal distributions

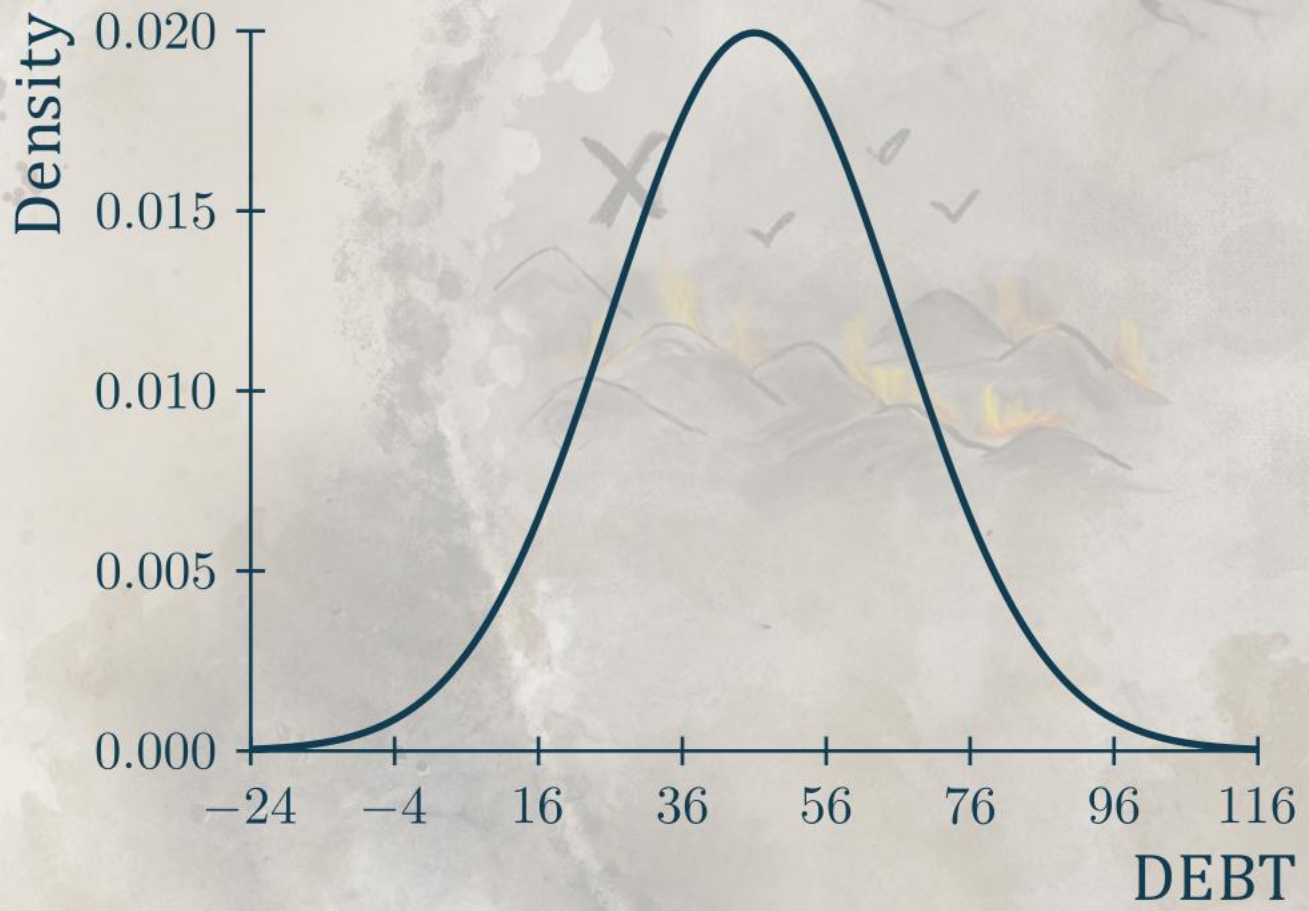
A priceless bounty

The typical pirate was a poor man



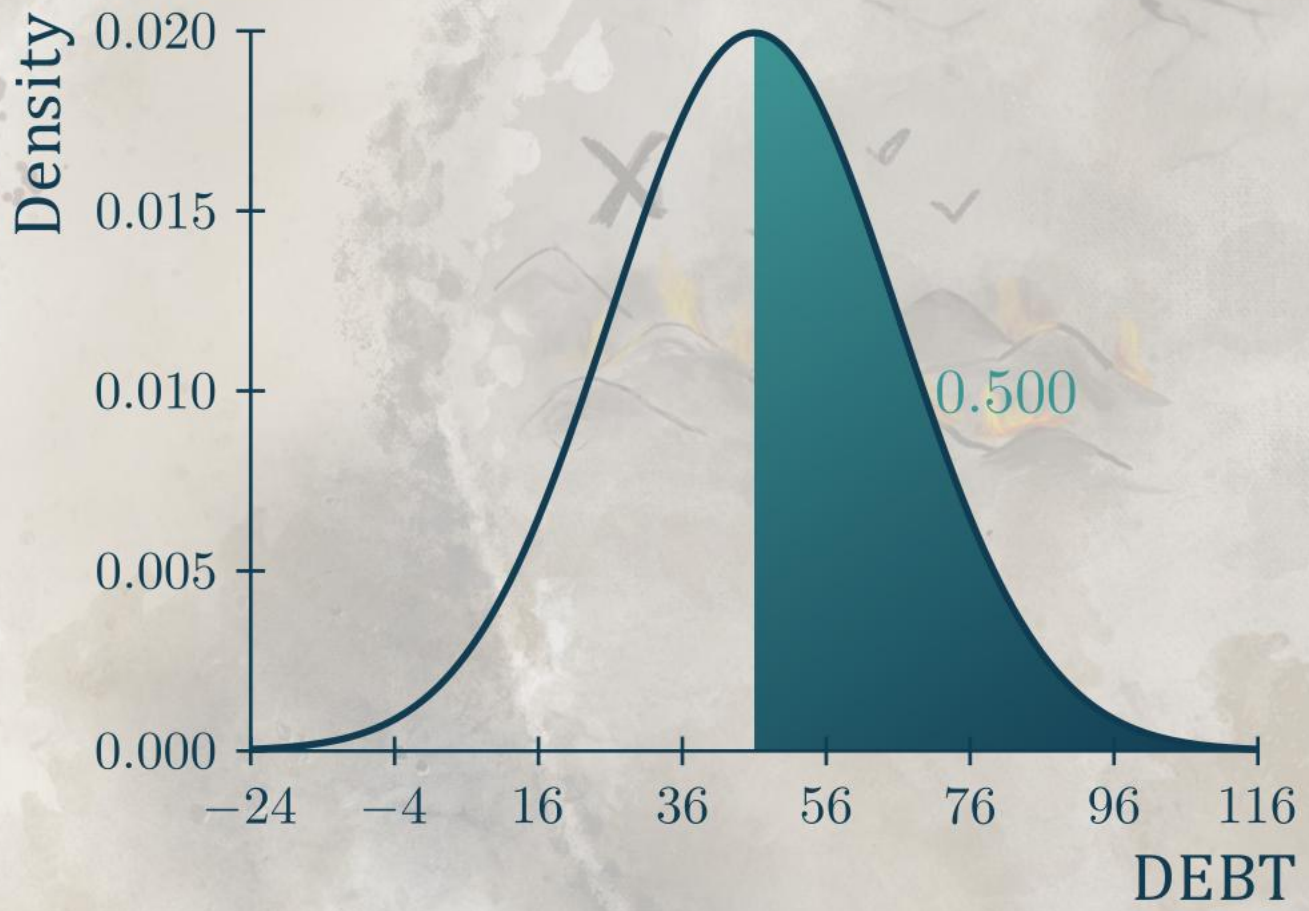
- And different swashbucklers owed people different amounts of money
- If we look at all pirates, their debts (in gold doubloons) followed a symmetric, unimodal distribution
  - Debt could also be negative, if one pirate owed another
- This distribution can be described pretty well with an *idealised* version: the so-called **normal distribution**

Smooth as an ocean wave



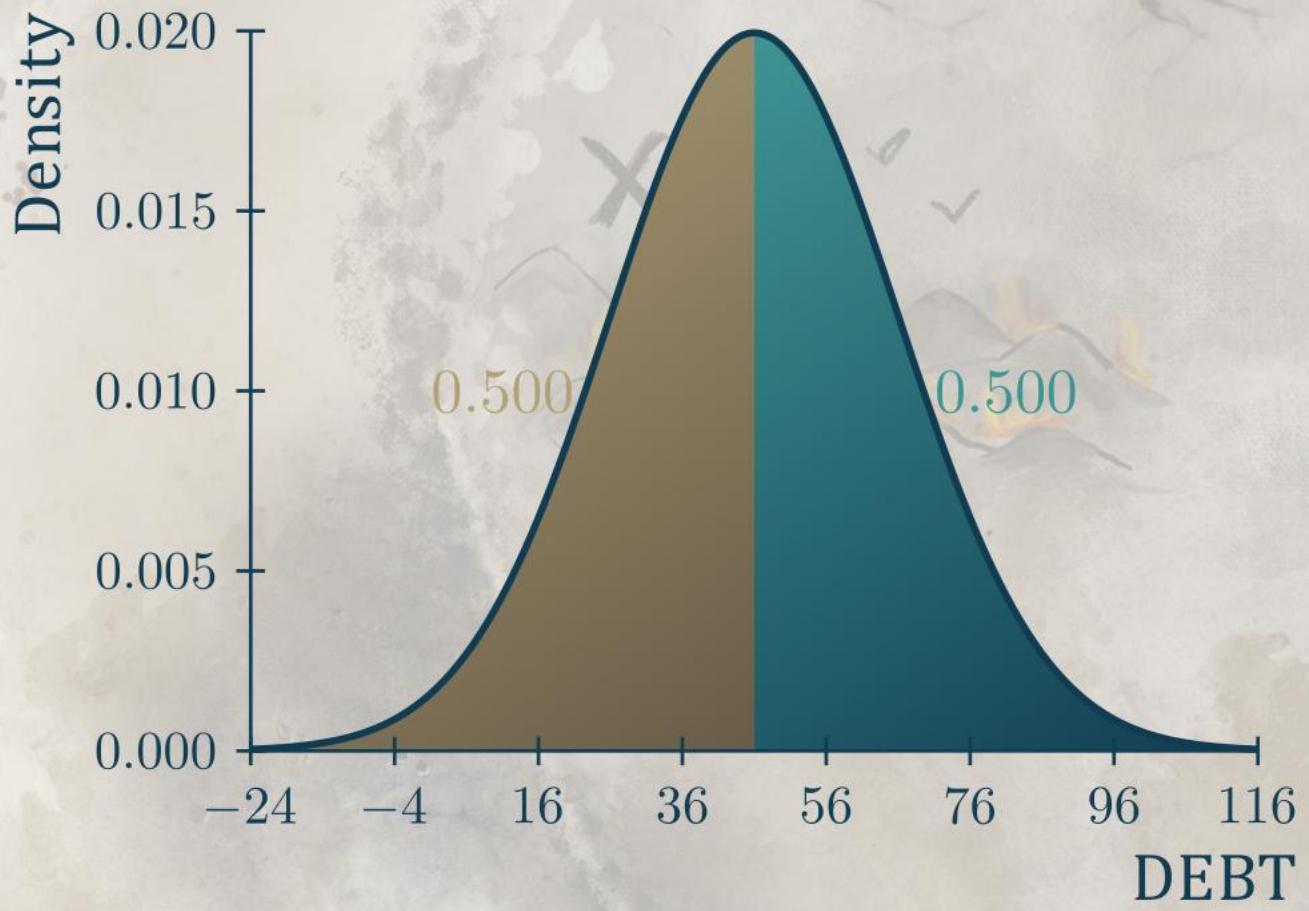
- The normal distribution is **continuous**: it assumes that the variable can have an infinite number of decimals
  - Not really the case with debt, but that's what we mean with *idealising* stuff. The normal distribution is just a **model** of reality
  - Variables that have 'gaps' in their scale (usually because they can only have whole numbers) are called **discrete** instead of continuous
- The normal distribution pops up in many places: in production processes, biological traits, stock indices, and so on
  - We even use it sometimes – just like today – to describe discrete variables…
  - … as long as we don't lose too much accuracy by pretending that they're continuous ☺

How to determine probabilities in a continuous distribution?



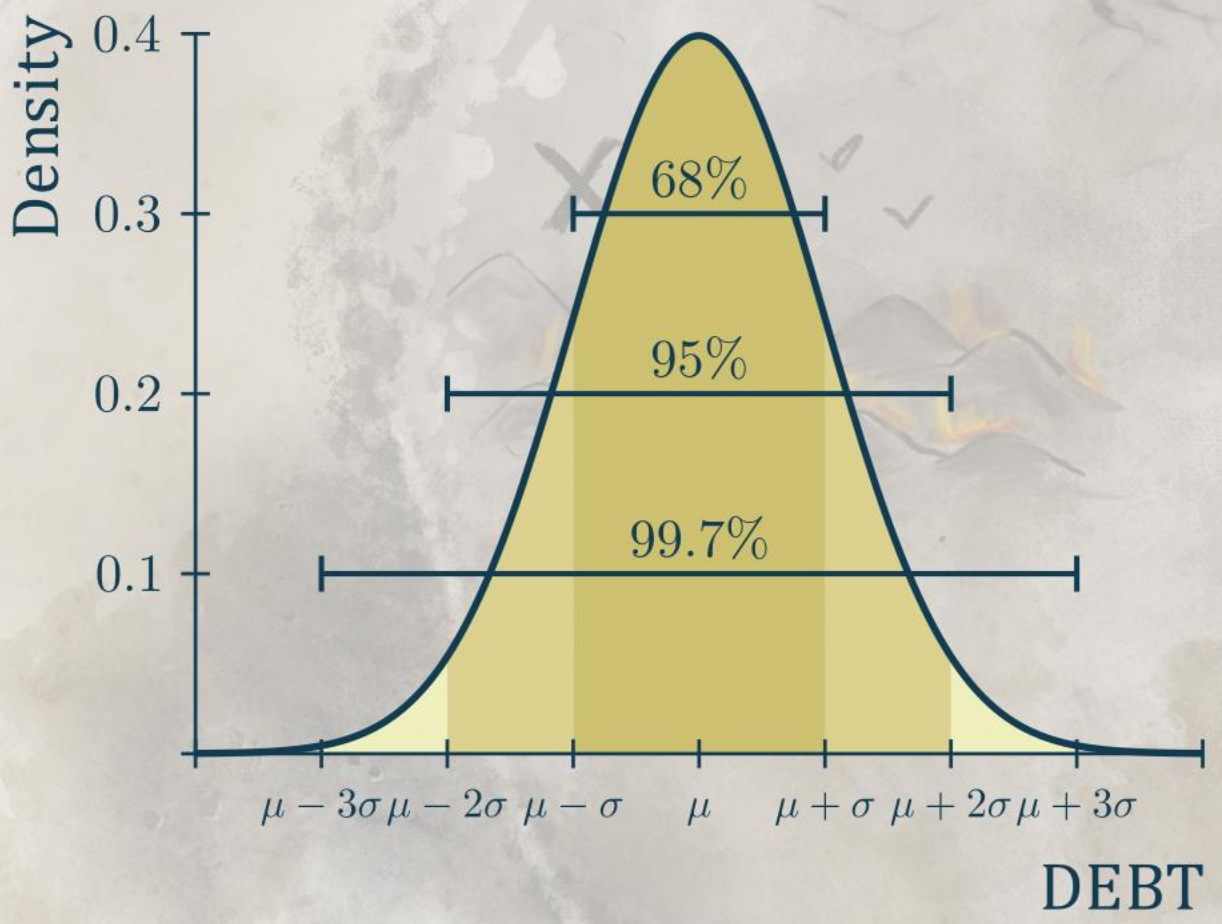
- The mean pirate's debt was $\mu = 46$ doubloons, with a standard deviation of $\sigma = 20$
- The total area under the curve is 1 (or 100%)
- Probabilities are <u>partial areas under the curve</u>
- Example: the probability that a random sailor would be *more than 46 doubloons* in the red is 50% (half of the probability area)

How to determine probabilities in a continuous distribution? (continued)



- So:
  - $P(x > 46) = 0.50$
  - $P(x < 46) = 0.50$
  - And $P(x = 46) = 0$
- Wait, what? Indeed: <u>exact</u> values have a probability of 0
  - $x = 46$ has no area under the curve (continuous variable!)
  - Only <u>intervals</u> have areas: 'more than 40 doubloons' or 'between 12 and 30'

Probabilities, method 1: rule of thumb



- 68% of the values are at most 1 standard deviation removed from the mean:

$$\mu \pm \sigma =$$
$$46 \pm 20 = [26, 66]$$

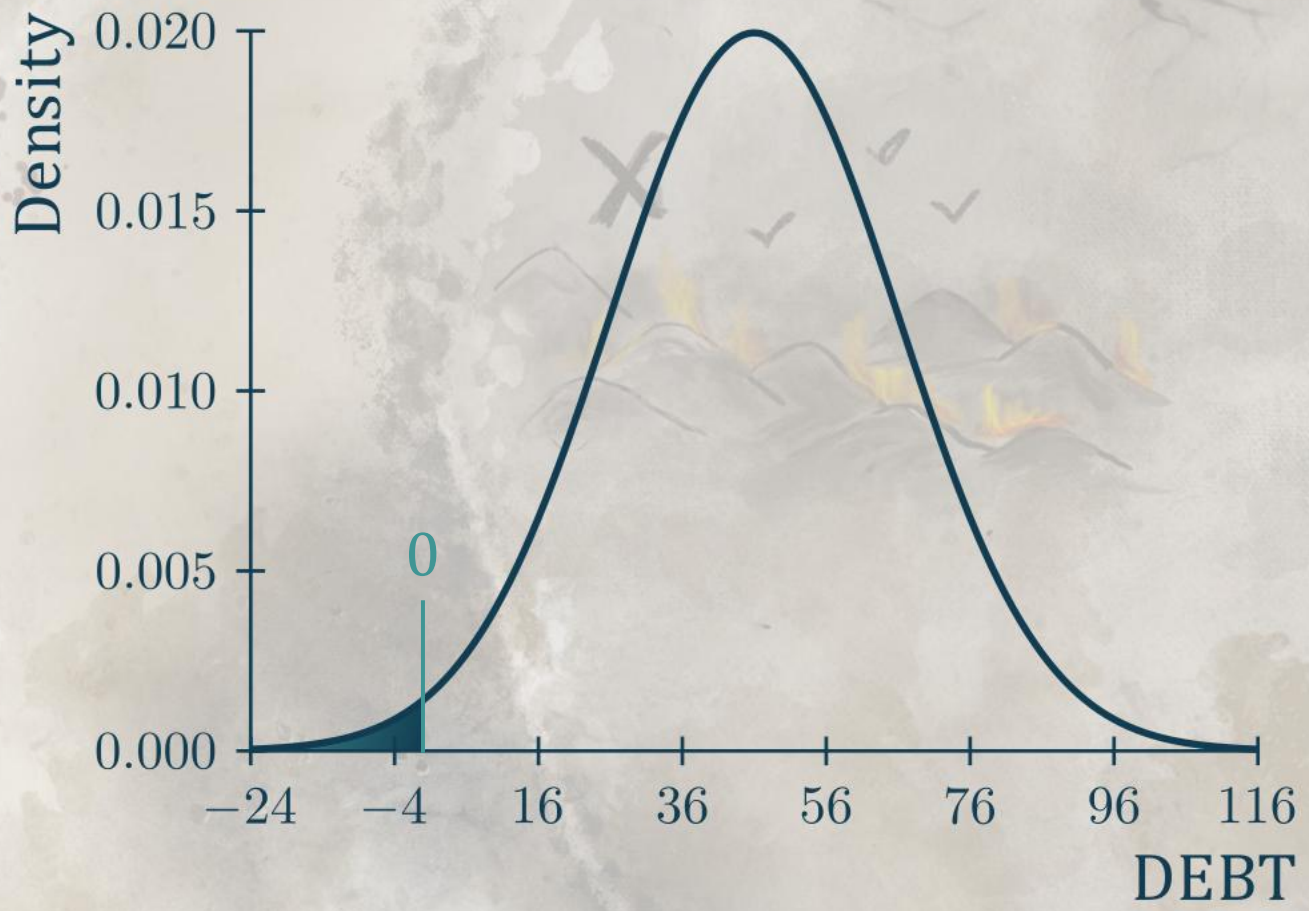
  - So if you met a random pirate on a deserted island, looking for treasure, there was a 68% chance that they were trying to pay off between 26 and 66 doubloons
- 95% lie no more than 2 standard deviations away from the mean:

$$\mu \pm 2\sigma =$$
$$46 \pm 40 = [6, 86]$$

- 99.7% (almost all) fall within 3 standard deviations from the mean:

$$\mu \pm 3\sigma =$$
$$46 \pm 60 = [-14, 106]$$

Probabilities, method 2: z-scores



- 'You owe me 4 doubloons' corresponds to saying 'You owe me 8 *escudos*'
  - They're two coins from the same Spanish currency system
  - So in a way, doubloons and *escudos* are different scales, but with the same meaning
  - One is a **re-expression** of the other
- The z-score is a similar kind of re-expression
- General formula:

$$z = \frac{y - \mu}{\sigma}$$

- Z-scores are handy in situations where the percentage rule won't work. Take a pirate whose finances were *neutral* (0 doubloons in debt):

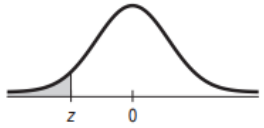
$$z = \frac{y - \mu}{\sigma} = \frac{0 - 46}{20} = \frac{-46}{20} = -2.3$$

- This buccaneer's debt would lie 2.3 *standard deviations below the mean*. Uncommon!
- What was the probability that you'd encounter such an individual? We can now use a lookup table

Using the z-table

Table Z

Areas under the standard Normal curve



	SECOND DECIMAL PLACE IN Z										
	0.09	*0.08*	*0.07*	*0.06*	*0.05*	*0.04*	*0.03*	*0.02*	*0.01*	*0.00*	*z*
										0.0000†	*−3.9*
	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	*−3.8*
	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	*−3.7*
	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.0002	*−3.6*
	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	*−3.5*
	0.0002	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	*−3.4*
	0.0003	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0005	0.0005	0.0005	*−3.3*
	0.0005	0.0005	0.0005	0.0006	0.0006	0.0006	0.0006	0.0006	0.0007	0.0007	*−3.2*
	0.0007	0.0007	0.0008	0.0008	0.0008	0.0008	0.0009	0.0009	0.0009	0.0010	*−3.1*
	0.0010	0.0010	0.0011	0.0011	0.0011	0.0012	0.0012	0.0013	0.0013	0.0013	*−3.0*
	0.0014	0.0014	0.0015	0.0015	0.0016	0.0016	0.0017	0.0018	0.0018	0.0019	*−2.9*
	0.0019	0.0020	0.0021	0.0021	0.0022	0.0023	0.0023	0.0024	0.0025	0.0026	*−2.8*
	0.0026	0.0027	0.0028	0.0029	0.0030	0.0031	0.0032	0.0033	0.0034	0.0035	*−2.7*
	0.0036	0.0037	0.0038	0.0039	0.0040	0.0041	0.0043	0.0044	0.0045	0.0047	*−2.6*
	0.0048	0.0049	0.0051	0.0052	0.0054	0.0055	0.0057	0.0059	0.0060	0.0062	*−2.5*
	0.0064	0.0066	0.0068	0.0069	0.0071	0.0073	0.0075	0.0078	0.0080	0.0082 *	*−2.4*
	0.0084	0.0087	0.0089	0.0091	0.0094	0.0096	0.0099	0.0102	0.0104	0.0107	*−2.3*
	0.0110	0.0113	0.0116	0.0119	0.0122	0.0125	0.0129	0.0132	0.0136	0.0139	*−2.2*
	0.0143	0.0146	0.0150	0.0154	0.0158	0.0162	0.0166	0.0170	0.0174	0.0179	*−2.1*
	0.0183	0.0188	0.0192	0.0197	0.0202	0.0207	0.0212	0.0217	0.0222	0.0228	*−2.0*
	0.0233	0.0239	0.0244	0.0250	0.0256	0.0262	0.0268	0.0274	0.0281	0.0287	*−1.9*
	0.0294	0.0301	0.0307	0.0314	0.0322	0.0329	0.0336	0.0344	0.0351	0.0359	*−1.8*
	0.0367	0.0375	0.0384	0.0392	0.0401	0.0409	0.0418	0.0427	0.0436	0.0446	*−1.7*
	0.0455	0.0465	0.0475	0.0485	0.0495	0.0505	0.0516	0.0526	0.0537	0.0548	*−1.6*
	0.0559	0.0571	0.0582	0.0594	0.0606	0.0618	0.0630	0.0643	0.0655	0.0668	*−1.5*

UNIT AND FIRST DECIMAL PLACE

- A bit old-fashioned, but a good way to make friends with the z-distribution ☺
- As it turns out*:

$$P(z \leq -2.30) = 0.0107$$

- So it was unlikely for a pirate to be free of debt: a chance of just 1.07%
- The probability that their debt was *higher* than 0 doubloons was therefore

$$P(z > -2.30) =$$
$$1 - P(z \leq -2.30) =$$
$$1 - 0.0107 = 0.9893$$

  - (This is the *complement rule*!)

The binomial distribution revisited

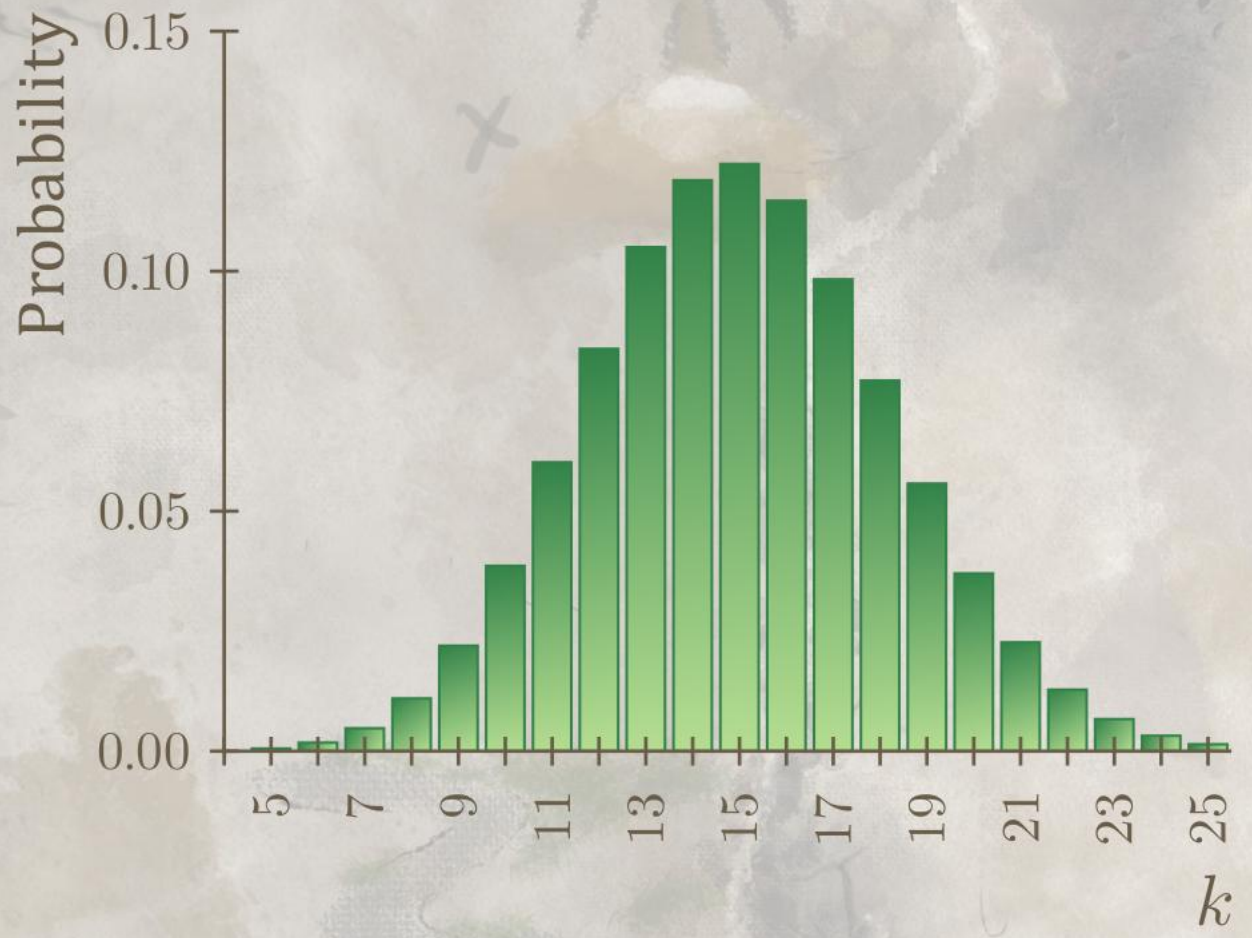
- In the late 1600s, the English Royal Navy cracked down on pirate vessels
- At the time, 30% of all ships were loaded with black market rum
- In June 1690, the English were on course to intercept n = 50 ships. How likely was it that at least a third (17) would carry illegal rum?

Catching rum smugglers

- We *could* use a binomial model…
  - n = 50 trials
  - What's the probability of 17 or more successes?

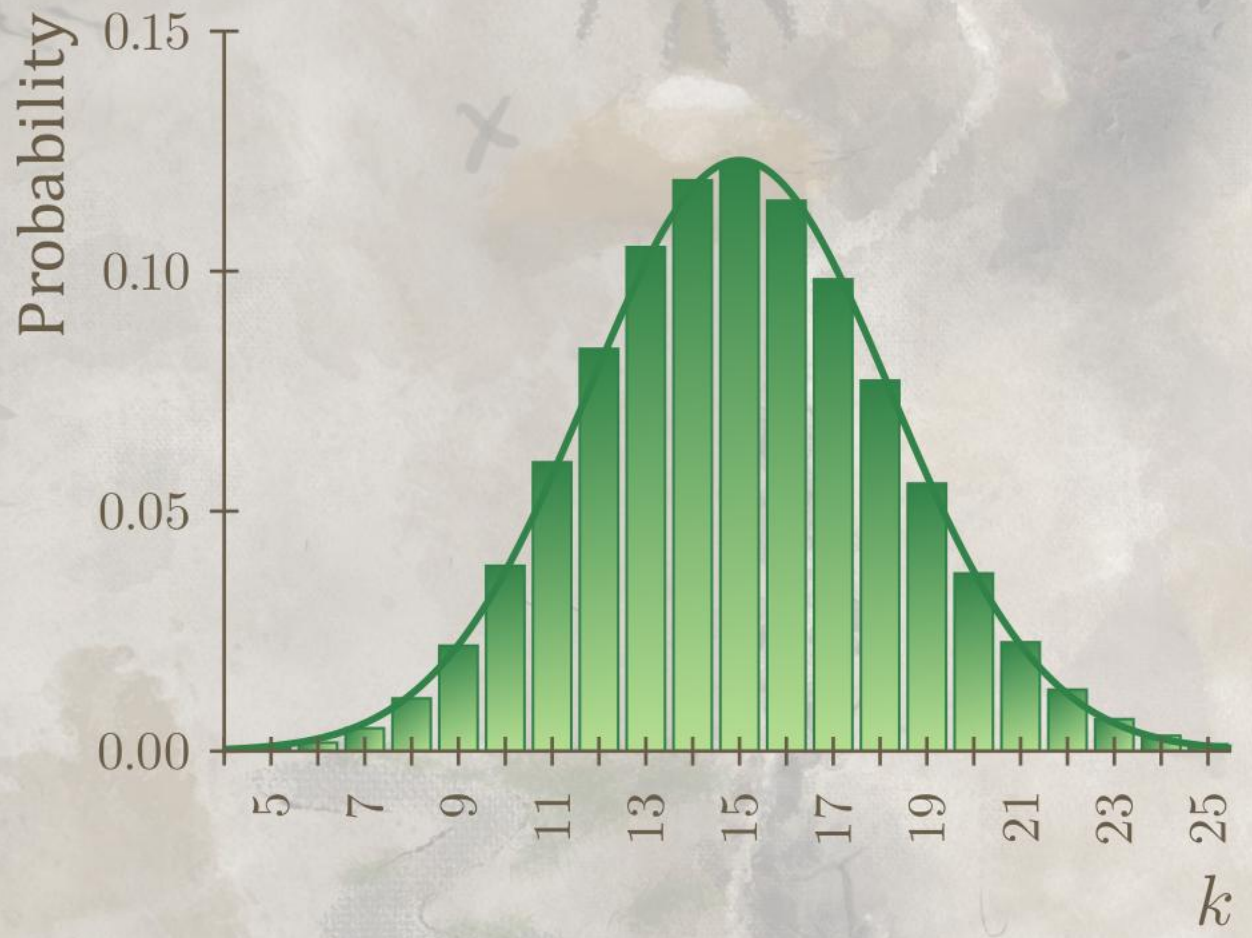
$$\boldsymbol{P(y \geq 17) =}$$
$$P(y = 17) +$$
$$P(y = 18) +$$
$$P(y = 19) +$$
$$\ldots +$$
$$P(y = 50)$$

- … Okay, no. That's way too much work 😐
  - (To be fair, computers have made it easier)



The normal approximation to the binomial

- But wait… this binomial distribution looks a lot like a <u>normal</u> one!
- **Success/Failure Condition:** *the binomial model is approximately normal if we expect at least 10 successes and at least 10 failures:*
  - $np \geq 10$ **and**
  - $nq \geq 10$
- Let's check. $p = 0.30$ and $q = 0.70$ so
  - $np = 50 * 0.30 = 15$ ☺
  - $nq = 50 * 0.70 = 35$ ☺
- Let us therefore just pretend (again) that we have a normal distribution, and apply the z-formula



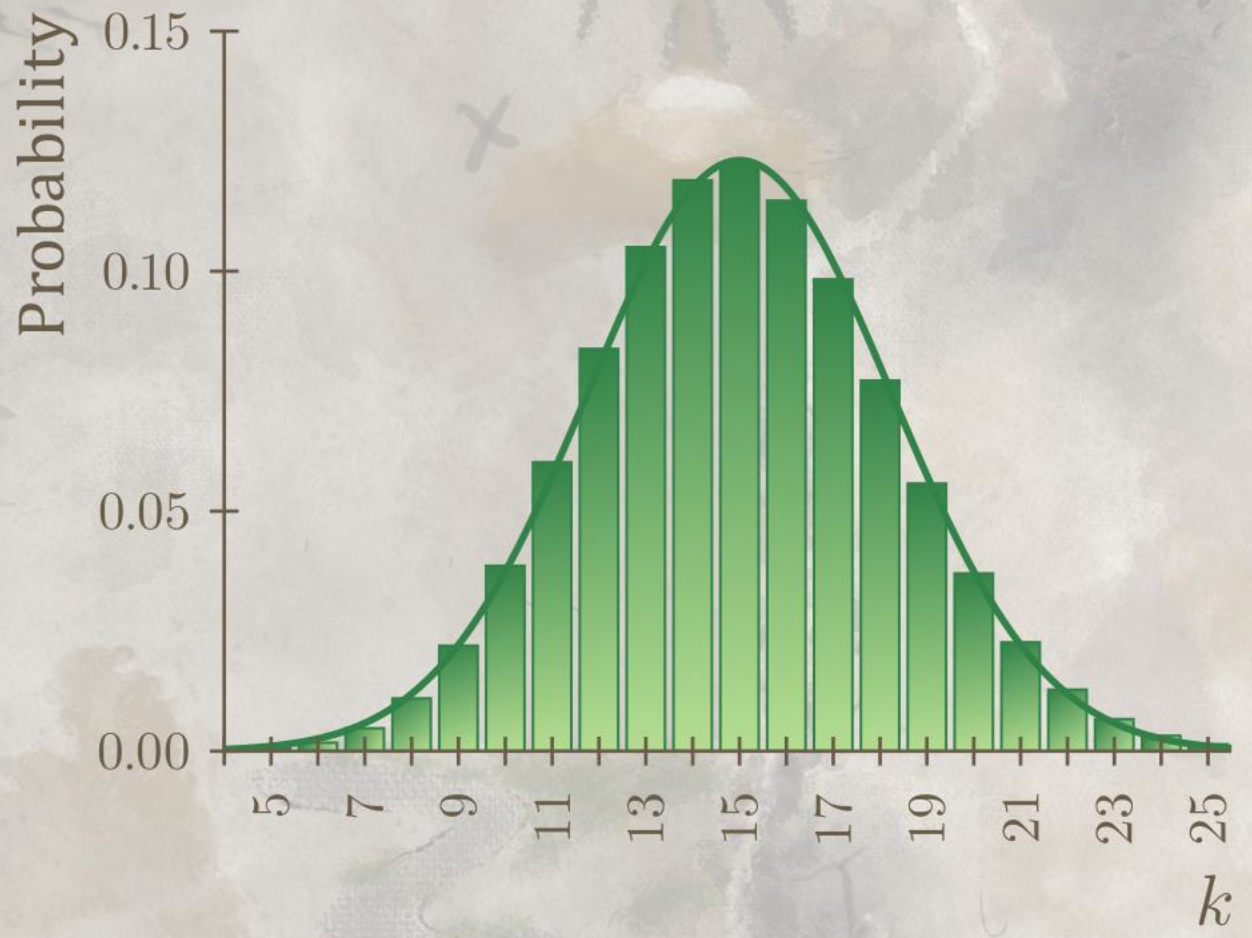
The normal approximation (continued)

- The parameters of this normal distribution are:
  - Expected value (mean) $\mu = np = 50 * 0.30 = 15$
  - Standard deviation $\sigma = \sqrt{npq} = \sqrt{50 * 0.30 * 0.70} = 3.24$
- We want the probability of finding at least 17 illegal rum traders, so

$$z = \frac{y - \mu}{\sigma} = \frac{y - np}{\sqrt{npq}} = \frac{17 - 15}{3.24} = 0.67$$

- The z-table tells us that

$$P(z \geq 0.67) = 1 - P(z < 0.67) = 1 - 0.7486 = 0.2514$$

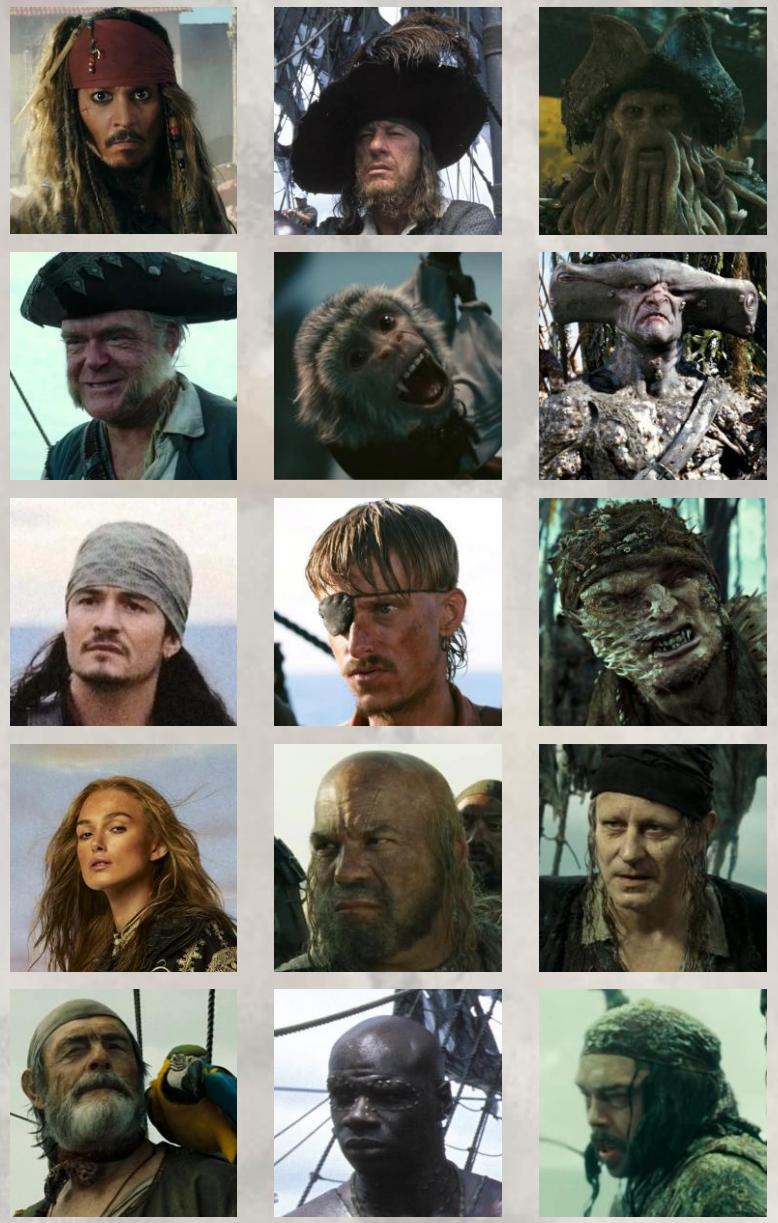


Sampling Designs

A motley crew

‘Yoho, yoho…
A pirate’s life for me!’

- Researcher Beckett (1698) wanted to conduct a study on job satisfaction among pirates
- For the sake of this lecture, imagine that the faces on the right constitute the population
- How can we come up with a representative sample?

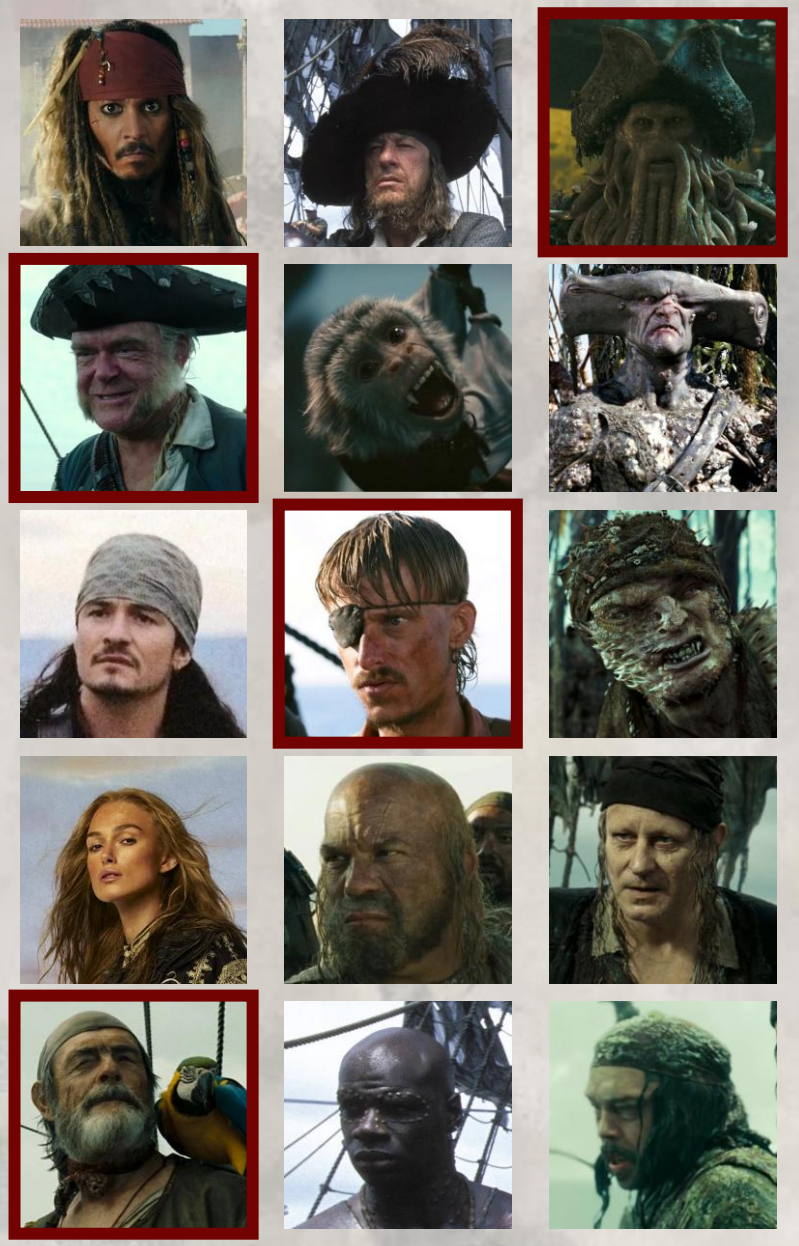


Simple random sample (SRS)

- Every individual has an equal (uniform) chance of getting drawn
- For all *calculations* we do in QM1 and QM2 from here on out, we'll assume that we draw an SRS

+ Super straightforward

– Effectiveness depends a lot on <u>sample size</u> *(n)*; small samples may end up unrepresentative due to bad luck (chance)



Stratified sample

- Identify pre-existing strata (a.k.a. subgroups) in the population, and draw a sample from each stratum
- Examples:
  - Different pirate ranks
  - Different age groups
  - Different genders
  - Different income groups
  - Different race groups (still a thing in the US)

captains

first mates

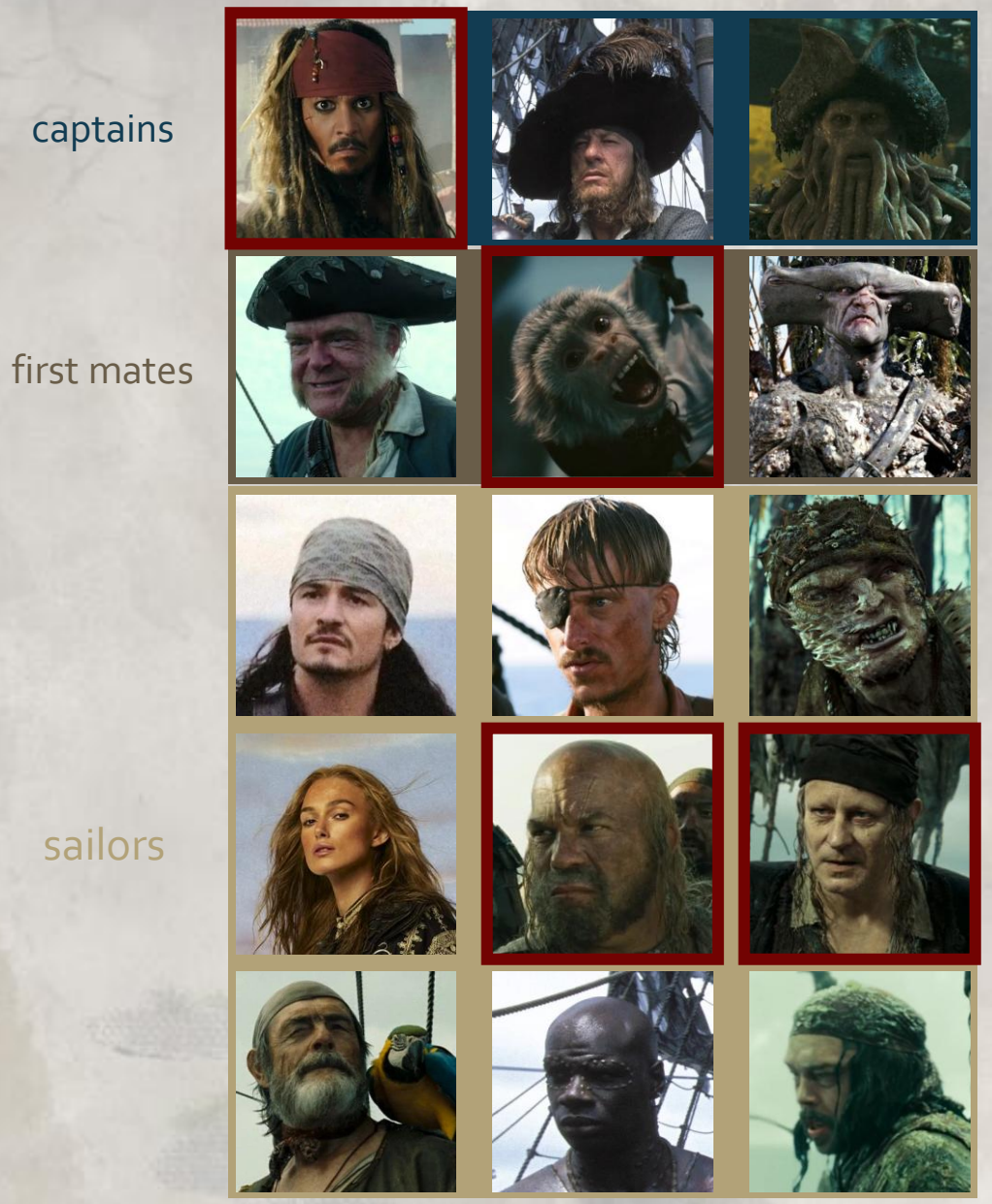
sailors

+ Every stratum is represented (if successful)

– Statistical calculations get more complicated (more than you think)

Stratified sample
(continued)

- So the idea of a stratified sample is:
  - You want take into account the variety in the population
  - So you sample from multiple strata that are clearly *different from each other* **(heterogeneous)**…
  - … but the members of each stratum are relatively *similar to each other* **(homogeneous)**



Cluster sample

- These pirates are from different ships, and every ship can be called a **cluster**: a pre-existing group (often by geographical location)
- In this case, wouldn't it be more practical to visit a single ship and have all crew members participate?
- This idea is exactly the opposite of stratified sampling:
  - You want take into account the variety in the population
  - So you sample one (or usually more) clusters. Different clusters are relatively *similar to each other* **(homogeneous)**…
  - … because they all contain individual members that that are clearly *different from each other* **(heterogeneous)**

crew #1 crew #2 crew #3

Cluster sample (continued)

- This way, the cluster(s) you take should make for a representative sample
- Other examples:
  - Sample a couple of hospitals and measure all patients in a particular clinic
  - Sample a couple of school classes from different schools, and measure all pupils in them
  - Sample a bunch of firms and measure all employees of each firm

+ Clusters often form a nicely varied set of people which mirrors the population

– Statistical calculations get more complicated (more than you think)

ship #1 ship #2 ship #3

Further topics for this week (self-study ☺)

- Sharpe, chapter 7
  - More calculation tricks with the z-table (check *In Practice 7.3* and *7.4*!)
- Sharpe, chapter 8
  - Types of observational studies
  - Populations versus samples
  - Multistage and systematic samples
  - How to sample badly (i.e. bias)

- Keep in mind: we expect you to read the whole book chapter. This slide highlights the most important aspects that were not covered in this lecture

Time fer some ship's biscuits!

After this: ye be learnin' mathematics
Until next week! ☺